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Model	Size	Hugging Face Identifier	Chat format
Llama 2	7B	meta-llama/Llama-2-7b-hf	–
	13B	meta-llama/Llama-2-13b-hf	–
	70B	meta-llama/Llama-2-70b-hf	–
Llama 2-Chat	7B	meta-llama/Llama-2-7b-chat-hf	✓
	13B	meta-llama/Llama-2-13b-chat-hf	✓
	70B	meta-llama/Llama-2-70b-chat-hf	✓
Code Llama	7B	codellama/CodeLlama-7b-hf	–
	13B	codellama/CodeLlama-13b-hf	–
	34B	codellama/CodeLlama-34b-hf	–
	70B	codellama/CodeLlama-70b-hf	–
Code Llama-Instruct	7B	codellama/CodeLlama-7b-Instruct-hf	✗
	13B	codellama/CodeLlama-13b-Instruct-hf	✗
	34B	codellama/CodeLlama-34b-Instruct-hf	✗
	70B	codellama/CodeLlama-70b-Instruct-hf	✗
DeepSeek	7B	deepseek-ai/deepseek-llm-7b-base	–
DeepSeek-Chat	7B	deepseek-ai/deepseek-llm-7b-chat	✓
DeepSeek-Coder	7B	deepseek-ai/deepseek-coder-7b-base-v1.5	–
DeepSeek-Coder-Instruct	7B	deepseek-ai/deepseek-coder-7b-instruct-v1.5	✗
DeepSeek-Math	7B	deepseek-ai/deepseek-math-7b-base	–
Gemma	8B	google/gemma-7b	–
Gemma-Instruct	8B	google/gemma-1.1-7b-it	✗
CodeGemma	8B	google/codegemma-7b	–
CodeGemma-Instruct	8B	google/codegemma-1.1-7b-it	✗
Llemma	7B	EleutherAI/llemma_7b	–
	34B	EleutherAI/llemma_34b	–
Llama	7B	huggyllama/llama-7b	–
FLoat	7B	nikhil07prakash/float-7b	–
Mistral	7B	mistralai/Mistral-7B-v0.1	–
OpenMathMistral	7B	nvidia/OpenMath-Mistral-7B-v0.1	–

Table 4: Details of all the models evaluated in the paper. The rightmost column indicates whether the chat format was used for prompting the model.

A Appendix

A.1 Model Details

We used the `transformers` library (Wolf et al., 2020) by Hugging Face for all our experiments. Table 4 presents all the models along with their Hugging Face identifiers.

For alignment-tuned models, we experimented with prompting the model with and without chat formatting (see Table 5 and Table 2 for the different formats). Based on results over a held-out development set, we selected the best-performing prompt format. We find that except for Llama 2-Chat models and DeepSeek-Chat, all other alignment-tuned models performed better with the non-chat format.

A.2 Constrained Decoding

```
Statement: Box 0 contains( [a-zA-Z]+)*, Box 1 contains( [a-zA-Z]+)*, Box
2 contains( [a-zA-Z]+)*, Box 3 contains( [a-zA-Z]+)*, Box 4 contains(
[a-zA-Z]+)*, Box 5 contains( [a-zA-Z]+)*, Box 6 contains( [a-zA-Z]+)*.
```

Figure 4: Regular expression used for constrained decoding of entity states.